2024 Apr 30

Vu H. N. Phan

Curriculum Vitae

Portland, Oregon
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Vu Phan is a Formal Verification engineer at Intel, performing pre-silicon validation of hardware designs. Vu earned a Ph.D. degree in Computer Science at Rice; his advisor was Prof. Moshe Vardi.

Experience

2022-now **Validation Engineer**, *Intel Corporation*.
- Perform pre-silicon Formal Verification of Integration Units in a System on a Chip

Skills: Computer Architecture, SystemVerilog, TCL

2017-2022 **Research Assistant**, *Rice University*.
- Designed and implemented algorithms to solve stochastic satisfiability with Prof. Moshe Vardi

Skills: Python, C++

2021 **Technical Intern**, *Intel Corporation*.
- Developed software to benchmark Formal Verification platforms

Certifications

2023 Mar **Jasper Formal Fundamentals v21.09**, *Cadence Design Systems*.

2021 Jul **SVA, Formal, and JasperGold Fundamentals for Designers v19.03**, *Cadence Design Systems*.

Education

2020-2022 **Doctor of Philosophy in Computer Science**, *Rice University*, GPA 3.63/4.00.

2017-2019 **Master of Science in Computer Science**, *Rice University*, GPA 3.60/4.00.

2014-2017 **Bachelor of Science in Computer Science & Math**, *Texas Tech University*, GPA 4.00/4.00.

2013-2014 **Associate of Science**, *Collin College*, GPA 4.00/4.00.

Projects

2019-2021 **DPMC**, *Dynamic-Programming Model Counter*, `https://github.com/vardigroup/DPMC`.
- Implemented a weighted-#SAT framework with Jeffrey Dudek

2018-2019 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, `https://github.com/vardigroup/ADDMC`.
- Implemented a weighted-#SAT solver

2016-2018 **LED**, *Language of Effective Definitions*, `https://vuphan314.github.io/LED`.
- Translated the literate-programming language `LED` into `SequenceL` and `LaTeX`

2015-2016 **L**, *Logic*, `https://github.com/iensen/LtoASPtranslator`.
- Translated the logic-programming language `L` into `Answer Set Prolog` with Evgenii Balai

Computing Courses

2017-2021 **Graduate Level**.

1. Reasoning about Software
2. Bioinformatics: Sequence Analysis
3. Statistical Machine Learning
4. Computer Systems Architecture
5. Multi-Core Computing
6. Compiler Construction
7. Artificial Intelligence
8. Programming Languages
9. Automated Program Verification
10. Logic in Computer Science

2015-2017 **Undergraduate Level**.

1. Operating Systems
2. Database Systems
3. Computer Architecture
4. Programming Languages
5. Software Engineering
6. Algorithms
7. Object-Oriented Programming
8. Computer Organization and Assembly Language
9. Modern Digital System Design
10. Data Structures
11. Automata
12. Programming Principles

Bibliography

Vu Phan is the correspondence author of the following publications (authors are sorted by surnames).

Theses

[Pha22] Vu H. N. Phan. "Quantitative Reasoning on Hybrid Formulas with Dynamic Programming". PhD thesis. Rice University, 2022. URL: `https://repository.rice.edu/items/2e464125-244d-431b-b998-612f0dc2b41a`.

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: `https://repository.rice.edu/items/a1a5e73d-a001-44ca-9730-25a7277c8af1`.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: `https://kasekopf.github.io/papers/sat21_procount.pdf`.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: `https://arxiv.org/abs/2008.08748`.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: `https://arxiv.org/abs/1907.05000`.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: `https://arxiv.org/abs/1809.09319`.